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## IN 2511 Computer Networks

### Lab Sheet 03

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#### **Objective:**

Study about IP address, subnet mask and default gateway and routing method (static or dynamic). You will be able to study IP addresses classes like A, B, C,D or E and how to assign subnet mask and ip addresses for given network according to given network address.

#### **ip address, subnet mask and Default gateway**

##### **IP Address**

An **Internet Protocol address (IP address)** is a numerical label assigned to each device (e.g., computer, printer) participating in a computer network that uses the Internet Protocol for communication. IP address has 32 bits numbers, and we are using IPv4 address. To identify computer or network device we have to use unique address is called IP address.

##### **Subnet mask**

Subnet mask is used to identify in range of IP address Net ID and the Host ID part.

A subnet mask is a number that defines a range of IP addresses that can be used in a network. Subnet masks are used to designate subnetworks, or subnets, which are typically local networks LANs that are connected to the Internet. Systems within the same subnet can communicate directly with each other, while systems on different subnets must communicate through a router. Therefore, subnetworks can be used to partition multiple networks and limit the traffic between them.

##### **Default gateway**

In computer networking, a **gateway** is a node (a router) on a TCP/IP network that serves as an access point to another network. A **default gateway** is the node on the computer network that the network software uses when an IP address does not match any other routes in the routing table.

##### **Static Routing**

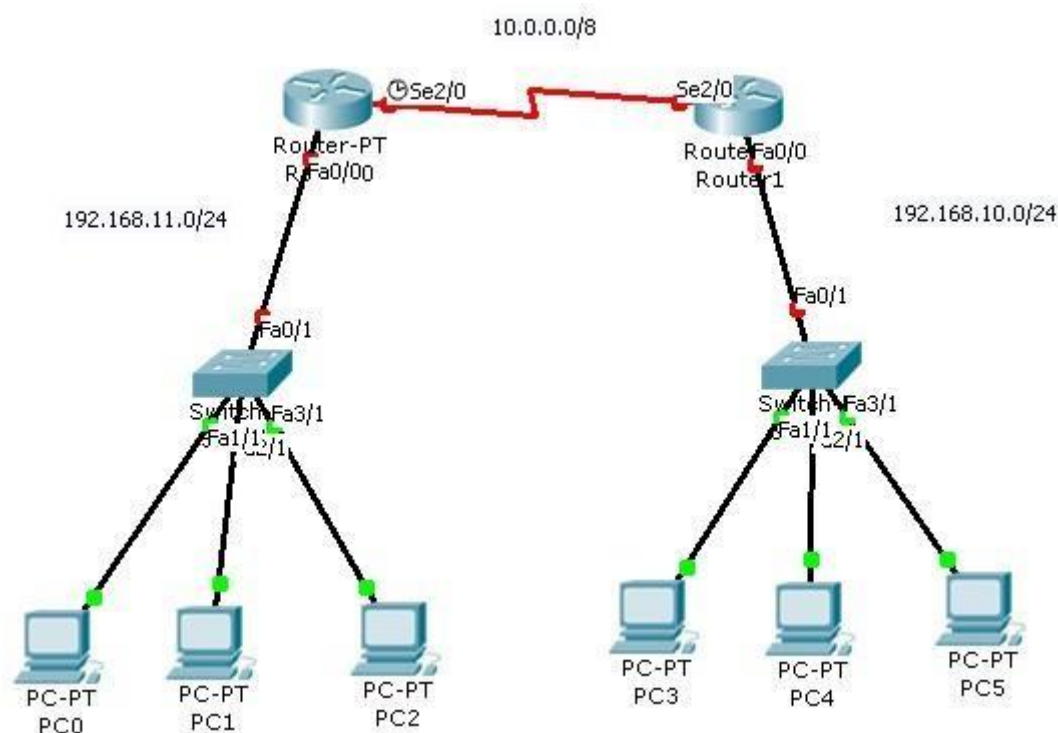
Static routing is simply the process of manually entering routes into a device's routing table via a configuration file that is loaded when the routing device starts up. As an alternative, these routes can be entered by a network administrator who configures the routes manually. Since these manually configured routes don't change after they are configured (unless a human changes them) they are called 'static' routes.

Static routing is the simplest form of routing, but it is a manual process. Static routing also does not handle failures in external networks well because any route that is configured manually must be updated or reconfigured manually to fix or repair any lost connectivity.

If you want to use static routing command you have to give below command for the routers in your network diagram.

**ip route <destination network address><subnet mask><next hop interface ip address>  
(not directly connected n/w)**

**Exercise 1:**



- 1) Design the above network using Packet Tracer simulator.
- 2) Assign suitable IP addresses for all the interfaces of all the routers and PCs of the above network.
- 3) After designing network try out send packet to other LAN using static routing.

**NOTE:** you can get the knowledge about how to design n/w, assign IP address according to the classes. You can try out giving different class mask IP address and see how packet are going in different LAN.